13 (a) Show that the equation $x^3 + 5x - 4 = 0$ has a solution between $x = 0$ and $x = 1$

Let $F(x) = x^3 + 5x - 4$.

$F(0) = -4$; $F(1) = 1 + 5 - 4 = 2$.

F is continuous and changes sign, so a root lies between 0 and 1.

1 mark
Evaluate endpoints

1 mark
Sign change; continuous

(2)

(b) Show that the equation $x^3 + 5x - 4 = 0$ can be arranged to give $x = \dfrac{4}{x^2 + 5}$

$x^3 + 5x = 4$

×	x^2	$+5$
x	x^3	$+5x$

$x(x^2 + 5) = 4$

$x = \dfrac{4}{x^2 + 5}$

1 mark
Rearrange and factor

1 mark
Divide by $x^2 + 5$

(2)

(c) Starting with $x_0 = 0$, use the iteration formula $x_{n+1} = \dfrac{4}{x_n^{\,2} + 5}$ twice,

to find an estimate for the solution of $x^3 + 5x - 4 = 0$

Start with $x = 0$.

First iterate $= \dfrac{4}{0^2 + 5} = 0.8$

Second iterate $= \dfrac{4}{0.8^2 + 5} = 0.709219858...$

1 mark
First value 0.8

1 mark
Substitute 0.8

After two iterations: $x \approx 0.709219858$.

1 mark
0.70921985...

(3)

(Total for Question 13 is 7 marks)